AP PRECALCULUS

Unit 1 • Lesson 1.6 • Guided notes

Topic 1.6 • Polynomial Functions and End Behavior

Guided notes • Fill in the blanks and work the examples in the boxes

Learning goals

- Describe end behavior with limit notation.
- Determine end behavior from the degree and the sign of the leading coefficient, in standard or factored form.
- Sketch a polynomial from its zeros, multiplicities and end behavior.

Vocabulary. end behavior, limit, $\lim_{x\to\infty}$, $\lim_{x\to-\infty}$, increases without bound, leading term.

1. Limit notation

$\lim\limits_{x\to\infty} p(x) = \infty$ means: as x _____, $p(x)$ _____.

$\lim\limits_{x\to-\infty} p(x) = -\infty$ means: as x _____, $p(x)$ _____.

2. The leading term controls the ends

Key idea

	$a_n > 0$	$a_n < 0$
n even	left _____, right _____	left _____, right _____
n odd	left _____, right _____	left _____, right _____

Example 1

Describe the end behavior of $p(x) = 3x^2 - 2x^5 + 7$.

Example 2 • Factored form

Describe the end behavior of $q(x) = -x(x-2)^2(3-x)$.

Example 3 • Sketching

Sketch $p(x) = (x+1)^2(x-2)$ using zeros, multiplicity, y-intercept and end behavior.

Common mistakes

- Using the first written term as the leading term.
- In factored form, missing a negative sign hidden in a factor like $(3-x)$.

Check your understanding

1. $\lim_{x\to-\infty}(4-x^3) =$ _____.
2. Both ends go down when the degree is _____ and the leading coefficient is _____.